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Coursre code:CSA0925-java programming

Assessment: CO4-AT2-lab practice

1.

Question 1 – Applet Life Cycle

Develop a Java Applet to demonstrate the Applet Life Cycle methods: init(), start(), paint(), stop(), and destroy().

Display a suitable message in the applet window whenever each life-cycle method is invoked. Observe the sequence of execution by running the applet and interacting with it.

Expected Outcome:

Students should understand the execution sequence and purpose of each Applet life-cycle method.

Output:



2.

Question 2 – Generating Graphic Images Using Applet

Develop a Java Applet that uses the Graphics class to generate a simple graphical scene.

The applet should draw:

A house

Sun

Trees

Road

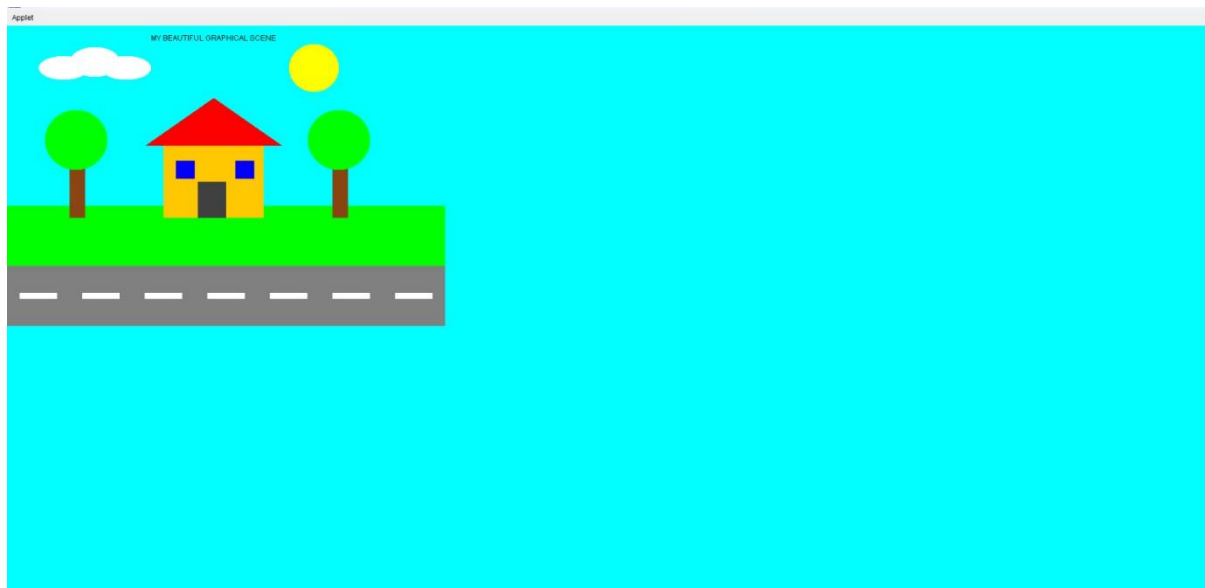
Clouds

Use methods such as drawLine(), drawRect(), drawOval(), fillRect(), fillOval(), and appropriate colors.

Expected Outcome:

Students should demonstrate the use of the Graphics class for drawing shapes and images.

Output:



3.

Question 3 – Calculator Application Using Applet Parameters

Develop a Calculator Applet that accepts two numbers and an arithmetic operator through HTML/Applet parameters.

The applet should perform:

Addition

Subtraction

Multiplication

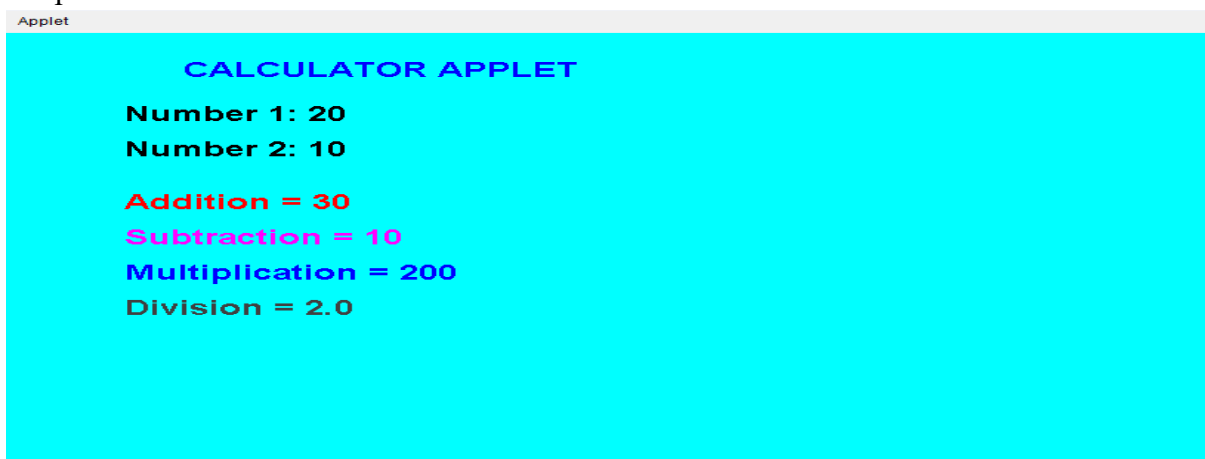
Division

Use the `getParameter()` method to retrieve the values passed to the applet.

Expected Outcome:

Students should understand how parameters are passed from an HTML page to an Applet and processed in Java.

Output:



4.

Question 4 – Student Profile Using AWT Components

Develop a Student Profile Application using Java AWT components such as:

Label
TextField
Checkbox
CheckboxGroup
Choice
TextArea
Button
List

The application should accept student details such as:

Name
Register Number
Gender
Department
Year
Skills/Programming Languages

When the Generate Report button is clicked, display the entered student information in a List or TextArea.

Expected Outcome:

Students should demonstrate the use of multiple AWT components and event handling.

Output:



5.

Question 5 – GridLayout with Event Handling

Develop a Java AWT application using GridLayout to create a simple form or calculator interface.

Arrange the components in a grid and implement appropriate event handling using ActionListener.

For example, create a calculator with buttons:

7 8 9 +
4 5 6 -
1 2 3 *
0 = C /

Perform the corresponding operation when a button is clicked.

Expected Outcome:

Students should understand GridLayout and button-based event handling.

Output:



6.

Question 6 – GridBagLayout with Event Handling

Develop a Java AWT Student Registration Form using GridBagLayout.

The form should contain:

Name

Register Number

Department

Email

Gender

Year

Submit Button

Reset Button

Use GridBagConstraints to properly position and align the components.

Implement event handling for the Submit and Reset buttons.

Expected Outcome:

Students should understand flexible component positioning using GridBagLayout and event handling.

Output:

The screenshot shows a window titled "Student Registration Form". It contains a registration form with the following fields and controls:

- Name:
- Register Number:
- Department:
- Email:
- Gender: ☒ Male ☐ Female
- Year:
- Buttons:

7.

Question 7 – BorderLayout with Event Handling

Develop a Java AWT application using BorderLayout.

Divide the window into:

NORTH – Header

SOUTH – Buttons

EAST – Options

WEST – Menu

CENTER – Display area

Add buttons in the interface and implement event handling so that clicking different buttons changes the content displayed in the center area.

Expected Outcome:

Students should understand the five regions of BorderLayout and event-driven programming.

Output:



8.

Question 8 – CardLayout with Event Handling

Develop a Java AWT application using CardLayout to create a multi-page interface.

Create at least three cards:

Card 1: Student Information

Card 2: Academic Information

Card 3: Confirmation/Report

Provide Next, Previous, and Submit buttons to navigate between the cards.

Implement event handling to switch between cards using CardLayout.

Expected Outcome:

Students should understand how CardLayout can be used to create multiple screens within a single window.

Lab Practice Evaluation Focus

Students will be evaluated based on:

Correct implementation of the required Java concepts.

Proper use of AWT components.

Correct implementation of event handling.

Appropriate use of layout managers.

User-friendly interface design.

Successful compilation and execution.

Ability to explain the program and output during the lab examination.

Output:

The application window is titled "Student Information System". It contains two cards that can be switched between.

Card 1: Student Information

Labels on the left: Name, Register Number, Program

Input fields on the right:

- Name: JSTH HANAN2024
- Register Number: 192472152
- Program: CSCI40

Buttons at the bottom: Previous, Next, Submit

Card 2: Academic Information

Labels on the left: Marks, Year

Input fields on the right:

- Marks: 977
- Year: 2024

Buttons at the bottom: Previous, Next, Submit

Confirmation Report

Name: S.D. HADARSHI Reg No: 180172152 Department: CSE-AI Marks: 877